Abstract

In this thesis we investigate a general pattern for sampling from probability distributions via Markov chains, namely the stochastic localisation approach. We first properly develop the language with measure theory and conditional laws. We then take a look at two concrete algorithms based on it, a continuous space one and a discrete space one. In the former case, we introduce time-reversed diffusions and take good care in deriving why exactly its equations are the way they are, in the modern language of probability laws and Schwarz distributions. In the latter case, we look at how sampling is done by observing the rate of a Poisson point processes, and extend this (known) algorithm in a couple directions.

Own work declaration

This thesis is written entirely by me, Alexander Gudev. All used literature is cited in place. Wherever a large language model has been used in developing ideas, its output has been thoroughly thought over and has been rewritten manually from scratch in this text.

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Contains 6705 words.

I Introduction and background

A. The Sampling Problem

or, *making something out of nothing*

The *sampling problem* is generally the problem of producing a sample from a probability distribution. In this thesis we develop the theory around a couple sampling algorithms based on the stochastic localisation approach.

And if the term "sampling from a probability distribution" sounds boring, let us briefly mention that this is just mathspeak for "generating [cool stuff]", where [cool stuff] stands for anything from from images and audio to weather and financial data, i.e. any type of objects that can be modelled as living in (a subset of) $\mathbb{R}^N$ and the likelihood of each instance thereof occurring is a well-defined probability distribution.

Depending on what is known, more specific sampling problems can be considered. Our base reference paper, (Montanari, 2025), outlines three such classes: we either "know" the target distribution or not, and if we "know" it, it might be only up to a parameter.

Task: **produce a sample** x from a distribution μ where

	we know	but do not know
Case I.	μ the distribution	
Case II.	$(\mu_\theta)_\theta$ and $(x_i)_i \overset{\perp}{\sim} \mu_\theta$ a family of distributions and a sample i.i.d. sequence	the parameter θ, and hence which μ_θ is the true one
Case III.	$(x_i)_i \overset{\perp}{\sim} \mu$ a sample sequence only	the distribution μ

A sampling algorithm is then packaged in different ways depending on which case we fall in: if μ is known, the algorithm is run directly, otherwise μ has to be estimated first.

Being an object of shape[1] $\mathcal{B}(\mathbb{R}^N) \to \mathbb{R}$ (so really a point in an infinitely dimensional space), a probability distribution is hard to estimate directly. We know, however, how to approximate functions of shape $\mathbb{R}^N \to \mathbb{R}$ via neural networks, and fortunately it will turn out that this is sufficient — a sampling algorithm won't need to know the full $\mu : \mathcal{B}(\mathbb{R}^N) \to \mathbb{R}$.

Reduction to a Score

Instead, the sampling algorithms we discuss here will depend on the target distribution μ only through a real-valued function on (finitely many) real arguments, $m_\mu(t, y_1, ..., y_n)$, so of shape

$$m_\mu : \mathbb{R}^{n+1} \to \mathbb{R},$$

[1] $\mathcal{B}(\mathbb{R}^N)$ denotes the set of $\mathcal{B}$orel subsets of $\mathbb{R}^N$.

in some contexts called a *score*, that we will define later (where?).

This means that to implement the algorithm, it suffices get a hold on m_μ. This indirectness of the algorithm's dependence on μ allows us to work even in cases where μ is not known (but assumed to exist), as then m_μ will be estimated from a dataset, e.g. by means of an artificial neural network.

So in this article, instead of the standalone problem of constructing a sampling algorithm for a given distribution μ, we consider the reduction

$$(\star) \qquad \begin{array}{c}\text{an estimator}\\ \text{of a particular}\\ \text{function } m_\mu\end{array} \quad \mapsto \quad \begin{array}{c}\text{a sampling}\\ \text{algorithm for } \mu\\ \text{based on } m_\mu\end{array} \quad \mapsto \quad x.$$

The prototype of such a reduction technique was proposed in (Jascha Sohl-Dickstein et al., 2015), where the sampling algorithm runs a Markov chain whose transitions are learned from data. A more particular instance of this technique, though, comes from using *stochastic localisations* as sampling algorithms as described in (Montanari, 2025).

In our work we will *assume* that the respective estimator job has been done, and will only focus on the sampling algorithm that *uses* said estimator. In terms of the conceptual chain $(\star)$ illustrating the reduction from sampling to estimation, we only construct the middle link. We will not tackle estimation of m_μ — this is a non-trivial task on its own; problems arising there have been explored in practice for example in (Yang Song & Stefano Ermon, n.d.).

B. Priors, Posteriors and Empirical Bayes

We will use a few terms from statistics throughout this article of which we provide a quick overview in this section.

A probability space $(\Omega, \mathbb{P})$ is assumed, i.e. a set Ω and a measure $\mathbb{P} : \mathrm{dom}(\mathbb{P}) \to [0,1]$ satisfying $\mathbb{P}(\Omega) = 1$. We leave $\mathrm{dom}(\mathbb{P})$, the domain of $\mathbb{P}$, unnamed, and we call its elements *events*. That is, an *event* is a subset $A \subseteq \Omega$ which is also an element of $\mathrm{dom}(\mathbb{P})$.

B.1 For Events

Imagine two events $\mathrm{A}, \Lambda \subseteq \Omega$, where Λ is mysterious and A is observable, by which we mean that

- we **cannot** observe Λ directly and (potentially) do not know its probability (hence mysterious), but
- we **do know** the conditional probability of A given Λ, $\mathbb{P}(\mathrm{A}|\Lambda)$, or even more — we may know if A would occur if Λ were known to (not) have happened. That is, we know the dependence of A on Λ.

Now let us observe A (or its complement). We would be interested in $\mathbb{P}(\Lambda|\mathrm{A})$, that is, what do we learn about the mystery from the observable?

The Bayes formula relates $\mathbb{P}(\Lambda|\mathrm{A})$ to $\mathbb{P}(\mathrm{A}|\Lambda)$, but it requires knowledge of $\mathbb{P}(\Lambda)$:

$$\mathbb{P}(\Lambda|\mathrm{A}) = \frac{\mathbb{P}(\Lambda)}{\mathbb{P}(\mathrm{A})} \cdot \mathbb{P}(\mathrm{A}|\Lambda). \tag{Bayes}$$

Thus *in the context of information being revealed*, the terms **prior** and **posterior** are introduced:

prior	**posterior**
$\mathbb{P}(\Lambda)$	$\mathbb{P}(\Lambda\|\mathrm{A})$
the knowledge of Λ	the knowledge of Λ
before we look at the A.	*after* we've looked at A.

B.2 For Random Variables

Now we generalize from events $\mathrm{A}, \Lambda \subseteq \Omega$ to random variables $Y, \lambda : \Omega \to \mathbb{R}$. As above, we consider a context where

- we may **not know** the law $\mathbb{P}_\lambda$ of the λatent variable, but
- we **do know** the law $\mathbb{P}_{Y|\lambda}$ of Y given any value of λ.

Mathematically, at any realization of λ, $\mathbb{P}_{Y|\lambda}$ is a distribution on the values of Y., i.e. $\mathbb{P}_{Y|\lambda}$ is a λ-indexed family of probability measures. But this is exactly[2] the definition of a Markov kernel. Symbolically, a Markov kernel is an object of shape

$$\mathbb{P}_{Y|\lambda} : \mathcal{B} \times \mathbb{R} \to [0, 1]$$

that, conceptually speaking[3], functions like a map

$$(A, a) \mapsto \mathbb{P}(Y \in A \mid \lambda = a).$$

What we're interested in now is the converse — $\mathbb{P}_{\lambda|Y}$, the conditional law of the λatent λ *given* the observable Y. And again we can use the terms *prior* and *posterior*, this time referring to the respective laws:

prior	**posterior**
$\mathbb{P}_\lambda$	$\mathbb{P}_{\lambda\|Y}$
the knowledge of λ	the knowledge of λ
before we observe Y.	*after* we've looked at Y.

Again, Bayes' formula gives the answer, at least in the discrete and non-degenerate case,

$$\mathbb{P}(\lambda = a \mid Y = y) = \frac{\mathbb{P}(\lambda = a)}{\mathbb{P}(Y = y)} \mathbb{P}(Y = y \mid \lambda = a),$$

but this is useful only if the prior, λ's law, is known so that we can plug in $\mathbb{P}(\lambda = a)$.

Statistics done under an assumption for the prior $\mathbb{P}_\lambda$ is thus called Bayesian, while for our sampling goals we take another route.

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[2] well, not *exactly*. An additional property is required that for all measurable A within the range of Y, the map $\lambda \mapsto \mathbb{P}(Y \in A|\lambda)$ is a measurable. We take the liberty of calling it *technical* and omit it from the main storyline.

[3] Technically we cannot write the conditional law that way because $\mathbb{P}(A|B)$ is defined only for conditions B of non-zero probability, so that the definition $\mathbb{P}(A|B) = P(AB)/P(B)$ makes sense. If λ takes values in $\mathbb{R}$, $(\lambda = a)$ may very well be an event of zero probability.

B.3 Empirical Bayesian Philosophy

Later we will define processes which allow us to learn about a λatent variable λ. Under the Bayesian philosophy, the law of λ is stipulated to be known, and then $\mathbb{P}(\lambda \mid A)$ is evaluated from Bayes' formula.

The stochastic localisation approach to sampling, though, will not rely on knowledge of the latent variable's distribution $\mathbb{P}_\lambda$, and so will instead employ the Empirical Bayes philosophy, under which $\mathbb{P}_\lambda$ is gradually estimated from observations as they come in.

(TODO: ADD LINKS TO ROBBINS' TALKS)

C. Unstarted Markov Processes

It is said that to move forward, one should first know their roots. In this first chapter, we will do exactly the opposite, and delete the past all up to the present. This thesis is all about going back.

> **Context.** We will live in a measurable space $(\Omega, \mathcal{F})$, see the world on a $\mathcal{B}$orel screen $D \subseteq \mathbb{R}^N$ and our clocks will tick with values in a set $I \subset [0, \infty)$ (so I or D can be discrete as well, e.g. $I = \mathbb{N}$ or $D = \mathbb{N}$).
>
> Additionally, our luck will be controlled by a probability measure $\mathbb{P} : \mathcal{F} \to [0, 1]$.

But even before having a probability measure, we can have a

> **Definition.** A **stochastic process** is an I-indexed sequence of random variables with values in D, i.e. a map $X : I \times \Omega \to D$ where $X(t, \cdot)$ is measurable at each fixed time $t \in I$.

A stochastic process defines what *can* happen. The law of X_t denoted by $\mathbb{P}_{X_t}$ then defines what is *likely* to happen.

This setup already fixes a starting position — the random variable X_0 with its law $\mathbb{P}_{X_0}$. As we will want to forget that, we focus our attention to processes with erasable past:

> **Definition.** A *Markov chain* is a stochastic process X satisfying the simple Markov property:
>
> $$\mathbb{P}\Big(X_{t_{k+1}} \in A \;\Big|\; X_{t_1} \in A_1, ..., X_{t_k} \in A_k\Big) = \mathbb{P}\Big(X_{t_{k+1}} \in A \;\Big|\; X_{t_k} \in A_k\Big).$$
>
> for all times $t_1 < ... < t_{k+1} \in I$ and $\mathcal{B}$orel $A, A_1, ..., A_k \subseteq D$.

The philosophy being that knowing the past of X (at points $t_1 ... t_k$) does not reveal any more information about the future of X, at any t_{k+1}, than its present at t_k.

In a sense, this is a *negative* definition: it tells how to filter out Markov processes from general processes. For our purposes, though, we need a *positive* definition, which tells what Markov chains *are*, not what they are not.

A Markov process induces[4] what is called a *kernel*[5]

$$\text{for any } s \leq t \in I, \quad \begin{aligned} P_{s,t} &: D \times \mathcal{B} \to [0,1] \\ P_{s,t}(x, A) &:= \mathbb{P}(X_t \in A \mid X_s = x), \end{aligned}$$

but a kernel on its own does not determine a stochastic process, not at least until a *starting point* (or distribution) is specified — different processes started at different points may share the same kernel (dynamics). In discrete time and space, it is known that a transition matrix with a choice of a starting point induces a Markov chain.

The kernels $\mathbb{P}(X_t \in A \mid X_s = x)$ describe X's evolution when started from almost any possible point $x \in D$. In that way, they gather X's present state + future behaviour, without any reference to the past.

Conceptually then we can decompose a Markov process $(X_t)_t$ into two orthogonal pieces:

$$\begin{matrix} \text{A Markov} \\ \text{chain } (X_t)_t \end{matrix} = \begin{matrix} \text{an initial} \\ \text{condition } X_0 \end{matrix} \oplus \begin{matrix} \text{all its Markov} \\ \text{kernels, } P_{s,t}. \end{matrix}$$

Whether such a decomposition formally holds for general Markov chains is out of our scope, and instead we focus on Itô processes.

Unstarted Itô Processes

> **Context.** Let time run smoothly $I = \mathbb{R}_{\geq 0}$ now and let a Wiener process $(W_t)_t$ land on our probability space.
>
> Pick also functions $b, \sigma : I \times D \to D$ satisfying the usual linear growth and local Lipschitz conditions to guarantee existence of solutions to SDEs.

An **Itô process** is a solution of a stochastic differential equation of the form

$$\begin{aligned} dX_t &= b(t, X_t)dt + \sigma(t, X_t)dW_t \\ X_0 &= x_0, \end{aligned}$$

which is defined to mean that the following equality holds:

$$X_t = x_0 + \int_0^t b(s, X_s)ds + \int_0^t \sigma(s, X_s)dW_s.$$

While the equation $dX_t = bdt + \sigma dW_t$ plus the starting point x_0 are what determine the solution, and we are used to saying that the expression $dX_t = b(t, X)dt + \sigma(t, X)dW_t$ is meaningless on its own, it is useful to treat the two as conceptually standalone pieces of the definition — akin to how the Markov kernels are standalone objects paired with each Markov chain.

After all, the same b, σ induce different solutions for different starting points x_0. So what does a standalone expression

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t$$

[4] We define it properly later.

[5] Called a *transition matrix* if D is discrete, $I = \mathbb{N}$ and $t = s + 1$.

mean? It is a map

$$x \mapsto \begin{pmatrix} \text{a solution } (X_t^{(x)})_t \text{ to} \\ \text{the SDE started at } x \end{pmatrix}.$$

Now, each solution $X^{(x_0)}$ on its own induces Markov kernels

$$x_0 \mapsto P_{s,t}^{(x_0)}(x, A) = \mathbb{P}\Big(X_t^{(x_0)} \in A \;\Big|\; X_s^{(x_0)} = x\Big),$$

which we expect (but do not prove here) to actually *not depend* on x_0 (as long as $s > 0$), because the $|X_s^{(x_0)} = x)$ condition already contains as much information as previous history does about the future: essentially it *restarts* the process.

The point is, we can think of the laws of an Itô process in the Markov context as an independent pair of an initial condition and all its kernels. The latter is what we have in mind by the adjective "Unstarted" in this section's title.

D. Stochastic Localisation

> **Context.** Pick a probability space $(\Omega, \mathbb{P})$, a subset $D \subset \mathbb{R}^N$ and a probability distribution $\mu : \mathcal{B}(D) \to [0, 1]$.
>
> Let x be a random variable whose distribution μ we will want to sample, i.e. let
>
> $$x : \Omega \to D$$
>
> be such that[6] $x \sim \mu$.

The only thing about x we care is its distribution. Think of x as *the* sample that we want our algorithm to produce. In the literature x is called a latent variable because we assume that we can't directly *measure* it but instead infer it from other variables.

The stochastic localisation approach to sampling consists of approximating the target distribution μ (from which a priori we do not know how to sample), with a sequence of ("corrupt") distributions $(\mu_t)_t$, each of which *can* be algorithmically sampled from.

Here we will look at two characterizations, as given by (Montanari, 2025), but it's worth noting that these are not the only ones: (Bobby Shi et al., n.d.) provides an insightful overview of perspectives of this framework.

We begin with the more hands-on one of the two.

D.1 Def. 1: Observation Processes

The idea is to "reach" x by a sequence of random variables $(Y_t)_t$ that clear out some type of noise.

> **Definition.** An **observation process** of x is a sequence of $\mathbb{R}^N$-valued random variables $(Y_t : t \in \mathbb{R}_+)$ such that

[6] For a D-valued random variable x and a measure $\mu : \mathcal{B}(D) \to [0, \infty]$, the notation $x \sim \mu$ is the statement that x is distributed as μ, i.e. that for all $A \in \mathcal{B}(D)$ we have $\mathbb{P}(x \in A) = \mu(A)$.

1. any finite sequence of the form
$$x, Y_{t_n}, \ldots, Y_{t_1}$$
forms a Markov chain, for all $t_1 < \cdots < t_n$;
2. all the Y_t together determine x in the sense that for any $\mathcal{B}$orel $A \subseteq \mathbb{R}^N$, the following conditional probability takes only the values 0 and 1:
$$\mathbb{P}(x \in A \mid \{Y_t : t \geq 0\}) \in \{0, 1\}.$$

The **stochastic localisation** of x's distribution then is defined as the sequence of conditional laws
$$\mu_t(A) := \mathbb{P}(x \in A \mid \{Y_s : s \leq t\}).$$

Y_t is what reveals more information about x as time progresses, where the information being revealed is encoded in the conditional laws μ_t. We will discuss this type of objects in more detail in Section B., but first we also give the second characterization of a stochastic localisation.

D.2 Def. 2: Measure Martingales

It directly describes the relationship between all μ_t and μ, and as usually stated may appear to be very abstract because it uses the term *random measure*.

A word on random measures

A measure has shape $\mathcal{B} \to [0, \infty]$ for some σ-algebra $\mathcal{B}$ on some set[7] $D \subseteq \mathbb{R}^N$, and *random* means having an Ω dependence, so a *random measure* is an object of shape
$$\Omega \to (\mathcal{B} \to [0, \infty]).$$

Thus it may appear that a random measure is a kind of higher-order probability, as there are two randomness sources here — Ω and $\mathcal{B}$. By abstract nonsense (uncurrying) a random measure is naturally equivalent to a two-argument function,
$$\Omega \times \mathcal{B} \to [0, \infty]$$

As discussed in Section B.2, though, this is precisely the shape of Markov kernels (conditional laws) on Ω. The reason we'll look at the Markov kernels as random measures instead is that the process they form turns out to be a martingale. One should keep in mind that we still have just one source of randomness, Ω, and the $(\mathcal{B} \to [0, \infty])$ are just projections of the presupposed probability $\mathbb{P}$ on D.

Definition. Pick a random variable x with values in $\mathbb{R}^N$ and law μ.

A **stochastic localisation** on Ω of μ is a measure-valued martingale that converges to δ_x, i.e. a sequence $(\mu_t)_t$ of objects of shape

[7] we're suggestively using the symbol $\mathcal{B}$ for the σ-algebra as we're only interested in random measures on $\mathcal{B}$orel subsets of $\mathbb{R}^N$.

$$\mu_t : \Omega \times \mathcal{B}(\mathbb{R}^N) \to \mathbb{R}$$

such that the martingale property $\mathbb{E}[\mu_t \mid \mathcal{F}_s] = \mu_s$ holds (as an equality of measures, for all times $s \leq t$) and $\mu_t \to \delta_x$ (again as measures, i.e. weakly).

The two conditions imposed are to be understood weakly, i.e.:

- The "martingale property as measures" means that fixing the last argument in

$$\mu : \mathbb{R}_{\geq 0} \times \Omega \times \mathcal{B}(\mathbb{R}^N) \to \mathbb{R}$$

  makes for a martingale in the classical sense;
- The convergence of the conditional laws μ_t to the unconditional law μ is expressed by first lifting μ to the conditional law δ_x. The weak limit just means fixing both arguments Ω and $\mathcal{B}(\mathbb{R}^N)$ and then taking classical limits.

D.3 Equivalence Between Def. 1 and 2

The equivalence of the two definitions[8] is shown in (Montanari, 2025), but it suffices to say that the two properties in each correspond to each other:

- the Markov observation process Y_t corresponds to the measure-valued martingale $(\mu_t)_t$ via $\mu_t = \mathbb{P}(x|\{Y_t : t \geq 0\})$;
- the second property in the first definition corresponds to μ_t having δ_x as a limit.

We will investigate the language used here in more detail in the next few sections, but we first outline how stochastic localisation is used in sampling.

E. Two Ways to Sample With Localisation

E.1 Diffusion Time-Reversal

Diffusion based sampling is the approach of time-reversal of a diffusion process.

Let $x \sim \mu$ and let W be a Wiener process independent from x. The typical process used here is the only stationary Gaussian process: the Ornstein-Uhlenbeck process

$$\begin{aligned} dZ_t &= -Z_t dt + \sqrt{2} dW_t \\ Z_0 &= x. \end{aligned}$$

From $e^{-t} de^t Z_t = dZ + Z dt$ the solution is obtained

$$\begin{aligned} Z_t &= e^{-t} x + \sqrt{2} e^{-t} \int_0^t e^s \, \mathrm{d}W_t \\ &= e^{-t} x + \sqrt{1 - e^{-2t}} \, G_t, \quad (x, G_t) \sim \mu \otimes \mathcal{N}(0,1) \end{aligned}$$

where we rescaled the stochastic integral[9] to get a unit normal[10] $G_t \sim \mathcal{N}(0,1)$.

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[8] under the assumption that the σ-algebra that Ω is endowed with is countably generated, which will allow the measures μ_t to be "converted" to random variables Y_t

[9] i.e. we put the random variable $G_t := (1/2(e^{2t} - 1))^{-1/2} \int_0^t e^s \, \mathrm{d}W_s$.

[10] $\int_0^t e^s \, \mathrm{d}W_s$ is normal of variance $V = \int_0^t e^{2s} \, \mathrm{d}s = \frac{1}{2}(e^{2t} - 1)$, thus $G_t := V^{-1/2} \int_0^t e^s \, \mathrm{d}W_s$ is normal of variance 1.

The explicit form of the solution shows that Z begins as an $x = Z_0$ and then transforms exponentially fast into a normal variable G_t. Hence if we can construct the backward process of Z_t, we can recover x by starting with a unit normal variable at $t = \infty$. It turns out that Z_t's reverse process *is* in fact a diffusion process again.

In practice then a sample x is obtained by running the reverse of Z_t over $[0, T]$ for a fixed T. As much as we'd want to, we cannot completely prove reversibility of diffusions here; instead we prove the form of the reverse process by deriving the "equations of motion" of the laws of a diffusion process - the Fokker-Planck equations in weak form.

A high-level overview of this technique is again in (Montanari, 2025) where the reverse equation is written directly: the time-reversed Ornstein-Uhlenbeck turns out to be a time-rescaled version of $tx + W_t$, which is an observation process and so its laws induce a stochastic localisation. As a byproduct of the derivations we will present here, though, we also see the connection to a closely related approach to sampling via Langevin dynamics.

E.2 Sampling in Time by a Poisson Process

The second application of stochastic localisation to sampling that is of interest to us is based on a Poisson point process. A very brief overview is again in (Montanari, 2025), sec. 4.8.

The reason we find this one curious is that it builds a sample of a *real* random variable from a Markov chain of *discrete* state space. The *reality* of the sample is encoded in time, not in space.

The idea is to encode the sample value in the frequency of a Poisson point process. More specifically, take a non-negative real random variable x and a process $(Y_t)_t$ that conditionally on x is Poisson point process

$$(Y \mid x) \sim \mathrm{PPP}(x).$$

In CHAPTER-PPP we derive how Y can be run in practice, prove convergence to x and develop the sampling scheme further with a couple generalizations. Let us quickly explain here *why* a Poisson point process makes sense at all as a base for sampling (if it can be run as an algorithm).

Recall that a Poisson point process $(X_t)_t$ of parameter $\lambda \in \mathbb{R}$ is defined to model *ticks* (points) in time that on average occur λ times per unit time. As time t progresses to infinity, the total number of occurrences, X_t, is expected to be $\lambda\, t$. Loosely speaking, $\lambda = \lim_t X_t/t$.

Now, at (almost) each random outcome $\omega \in \Omega$, the target sample has value $x(\omega)$. So if we have a point process (with counting variable) $(X_t)_t$ of rate $x(\omega)$, we can expect to recover the rate $x(\omega)$ from $\lim X_t(\omega)/t$.

Of course, at individual outcomes $\omega \in \Omega$ in isolation we cannot talk about $(X_t(\omega))_t$ having rate $x(\omega)$. The way to define a point process of such a random rate is by *mixing* otherwise defined point processes: take for each $\lambda \in \mathbb{R}$ a (different) process $X^{(\lambda)} \sim \mathrm{PPP}(\lambda)$, then a process X is defined to be a process of random rate x if

$$\mathbb{P}(X \in A) = \int \mathbb{P}\big(X^{(\lambda)} \in A\big)\, \mathrm{d}\mathbb{P}(x = \lambda), \tag{1}$$

i.e. if the law of X is a convex combination of the laws of all $X^{(\lambda)}$ and the λ are taken from the distribution of x; in more classical notation the integral is written $\int \mathbb{P}\big(X^{(x)} \in A\big) d\mu_x$ with μ being the law of x.

II Stochastic localisation with time-reversed diffusions

Before we even attempt to reverse time, we need to understand how it flows forward.

A. What Is It That Evolves?

We want to construct a differential equation describing the evolution of the laws of Y_t. Inspired from PDEs on regular functions of the form $\partial_t p(t,x) = F[t, x, \partial_x, ...]$ in mind, we might be tempted to simply replace the function p by the law $\mathbb{P}|Y_t\rangle$ and develop an equation for $\partial_t \mathbb{P}|Y_t\rangle$ (if exists). This naïve way leads to inconsistencies that we outline below.

In particular, a well-behaved generalization of $\partial_t p(t,x)$ should be stable under changes to the initial conditions (the law of Y_0). We now show that $\partial_t \mathbb{P}|Y_t\rangle$ is not.

A.1 The Problem With $\partial_t \mathbb{P}|Y_t\rangle$

> **Context.** For two random variables ξ, η and a real (rate) $r > 0$ construct the processes
>
> $$X_t = \xi + r\,t \quad \text{and} \quad Y_t = \eta + r\,t, \tag{2}$$
>
> satisfying the SDEs
>
> $$\begin{matrix} dX_t = r\,dt \\ X_0 = \xi \end{matrix} \quad \text{and} \quad \begin{matrix} dY_t = r\,dt \\ Y_0 = \eta \end{matrix}$$
>
> sharing the same dynamics $dX_t = r\,dt = dY_t$ but having different initial conditions ξ, η.

So we'd want our generalization of $\partial_t p$ to be the same on both processes.

Derivation. Evaluate $\partial_t \mathbb{P}|X_t\rangle|_{t=0}$ on $[a, \infty)$ for $a \in \mathbb{R}$:

$$\begin{aligned} \partial_t\, \mathbb{P}(X_t \geq a)|_{t=0} &= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon}(\mathbb{P}(X_\varepsilon \geq a) - \mathbb{P}(X_0 \geq a)) \\ &\overset{(2)}{=} \lim_{\varepsilon \to 0} \frac{1}{\varepsilon}(\mathbb{P}(\xi + r\varepsilon \geq a) - \mathbb{P}(\xi \geq a)) \end{aligned}$$

(express[11] $(\xi + r\varepsilon \geq a) = (\xi \geq a) \cup (a - r\varepsilon \leq \xi < a)$)

$$= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \mathbb{P}(a - r\varepsilon \leq \xi < a), \tag{3}$$

And similarly for Y:

$$\partial_t\, \mathbb{P}(Y_t \in [a, \infty))|_{t=0} = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \mathbb{P}(a - r\varepsilon \leq \eta < a).$$

[11] Having the usual notation for events in mind, $(\xi \geq a) \equiv \{\omega \in \Omega \mid \xi(\omega) \geq a\}$

It is evident now that the derivative depends directly on the laws of ξ and η. If the two were chosen according to, for example, $\xi \sim \text{Unif}(0,1)$ and $\eta \sim \text{Unif}(0.5,1)$, and we pick $a = 1$, the limits above would evaluate to

$$\begin{aligned} \partial_t \mathbb{P}(X_t \geq 1) &= r, \\ \partial_t \mathbb{P}(Y_t \geq 1) &= 2r, \end{aligned} \tag{4}$$

because in that case

$$\begin{aligned} &\mathbb{P}(1 - r\varepsilon \leq \xi < 1) = r\varepsilon \\ \text{but} \quad &\mathbb{P}(1 - r\varepsilon \leq \eta < 1) = 2r\varepsilon. \end{aligned}$$

□

Conclusion. the time derivative of the laws of X_t, Y_t is not invariant to change of initial conditions.

A.1.1 Why does $\partial_t \mathbb{P}|Z_t\rangle$ vary?

It turns out we can easily visualize this phenomenon if we consider

1. the speed at which (the laws of) X_t and Y_t move through $\mathbb{R}$, and,
2. the speed at which individual sets $A \subseteq \mathbb{R}$ gain or lose weight[12] as t varies, under $\mathbb{P}|X_t\rangle$ and $\mathbb{P}|Y_t\rangle$,

and realize that $\partial_t \mathbb{P}|X_t\rangle$ measures the latter, while the invariant quantity is the former.

In Figure 1, the highlighted areas indicate the weight gained by $[1, \infty)$ as time progresses from 0 on. While both X_t and Y_t, as well as their plotted densities, are moving to the right at the same rate r (both driven by a shared $dX = r\,dt = dY$), the rate of weight gain per unit time of $[1, \infty)$ is twice as large in the Y case as in the X case.

A.2 The Time-Derivative Of $\mathbb{P}|X_t\rangle$

> **Context.** Fix a process $Z_t = \zeta + r\,t$ where ζ is a random variable with CDF F_ζ and continuous density $p_\zeta = F'_\zeta$.

Derivation. Pick $a \in \mathbb{R}$ and evaluate the probability gain over $(-\infty, a)$ at t:

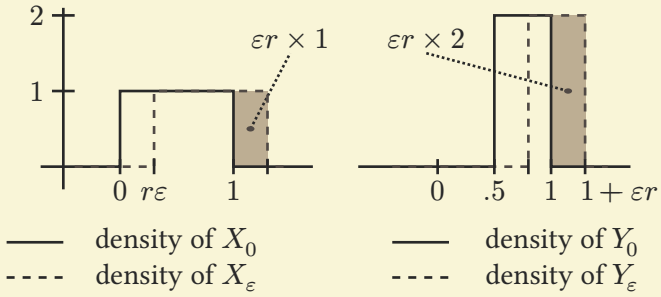


Figure 1: The densities of X and Y at times $t = 0$ and $t = \varepsilon$. Here again $\xi \sim \text{Unif}(0,1)$ and $\eta \sim \text{Unif}(0.5,1)$.

[12] measure

$$\begin{aligned}\mathbb{P}(Z_t \le a) - \mathbb{P}(Z_0 \le a) &= \mathbb{P}(\zeta + r\,t \le a) - \mathbb{P}(\zeta \le a)\\ &= \mathbb{P}(\zeta \le a - r\,t) - \mathbb{P}(\zeta \le a)\\ &= -\mathbb{P}(a - r\,t \le \zeta \le a)\\ &= -\int_{a-r\,t}^{a} p_\zeta(a)da \qquad (5)\end{aligned}$$

(by the mean value theorem, for some $a' \in (a - r\,t, a)$)

$$= -r\,t \times p_\zeta(a').$$

This gives us a Taylor expansion arount $t = 0$:

$$\mathbb{P}(Z_\varepsilon \le a) = \mathbb{P}(Z_0 \le a) - r\varepsilon \times p_\zeta(a) + o(\varepsilon^2). \qquad (6)$$

The term $r\varepsilon \times p_\zeta(a)$ corresponds to the highlighted rectangles in Figure 1, and where the invariant quantity r is "corrupted" by the multiplier $p_\zeta(a) = F'_\zeta(a)$.

Recall that we can recover the law of a real random variable from its CDF $F(a) = \mu(-\infty, a]$, by the Lebesgue-Stieltjes integral $\mu(A) = \int_A dF(a)$. Denote[13] the measure recovered this way by $\mathbb{P}|F\rangle$. Over all $a \in \mathbb{R}$, Equation 6 is an equation of CDFs, so it naturally induces an equation of *laws*. The (signed) measure induced by the first order term $rp_\zeta(a) = rF'_\zeta(a)$ is then $r\mathbb{P}\big|F'_\zeta\big\rangle$, so Equation 6 in measure speak reads

$$\mathbb{P}|Z_\varepsilon\rangle = \mathbb{P}|Z_0\rangle - \varepsilon r\,\mathbb{P}\big|F'_\zeta\big\rangle + o(\varepsilon^2).$$

To recover r when taking the derivative $\partial_t\mathbb{P}|Z_t\rangle$ now it is natural to also divide by $\mathbb{P}\big|F'_\zeta\big\rangle$ in Radon-Nikodym sense:

$$r = \frac{\mathrm{d}\partial_t\mathbb{P}|Z_t\rangle}{\mathrm{d}\mathbb{P}\big|F'_\zeta\big\rangle},$$

or, writing it in higher-order derivative notation,

$$r = \frac{\partial\,\mathrm{d}\mathbb{P}|Z_t\rangle}{\partial t\,\mathrm{d}\mathbb{P}\big|F'_\zeta\big\rangle}\bigg|_{t=0},$$

which indicates how from the increment of $\mathbb{P}|Z_t\rangle$ we cancel a time t and a density $\mathbb{P}\big|F'_\zeta\big\rangle$ factor. □

That is, we reach the following

Conclusion. A natural initial-distribution-invariant quantity based on the time derivative of Z_t's law is its Radon-Nikodym derivative against the signed measure induced by (the Lebesgue-Stieltjes integral with respect to) Z_0's density.

Two problems arise in the derivation of the last expression: for it to work,
- $Z_0 = \zeta$ must admit a (continuous) density $p_\zeta = F'_\zeta$;
- even then, the absolute continuity condition $\mathbb{P}\big|F'_\zeta\big\rangle \ll \mathbb{P}|Y_t\rangle$ must hold.

[13] No ambiguity can arise with the notation $\mathbb{P}|\zeta\rangle$ and $\mathbb{P}|F\rangle$ because ζ is of shape $\Omega \to \mathbb{R}^N$, while F is of shape $\mathbb{R} \to \mathbb{R}$. And we have $\mathbb{P}|\zeta\rangle = \mathbb{P}\big|F_\zeta\big\rangle$ if ζ is $\mathbb{R}$-valued.

Thus it is important to go back to Equation 6 and proceed in a way that does not require canceling a F'_ζ term.

A.3 Getting ~~Out Of~~ Into the Box

In Figure 1 we treated the increment $\mathbb{P}|Z_\varepsilon\rangle - \mathbb{P}|Z_0\rangle$ to the first order in ε as a rectangle. We then obtained its width by dividing its mass by its height: $\big(\varepsilon r \times p_\zeta\big)/p_\zeta$. Now, while ζ's density p_ζ might not exist, its mass always does.

And if we write $p_\zeta = 1 \times p_\zeta$, the quantity is numerically identical but probabilistically understandable as a unit weight of ζ, which we always can compute from $\mathbb{P}|\zeta\rangle$. Can we divide the area $\big(\varepsilon r \times p_\zeta\big)$ by $1 \times p_\zeta$ — a unit mass of ζ then, and recover r?

This would correspond to dividing the mass $\big(\varepsilon r \times p_\zeta\big)$ by the mass $\big(1 \times p_\zeta\big)$, thus arriving at a *dimensionless* quantity εr. But that is nonsensical, because εr has units of *space*:

$$Z_\varepsilon = \zeta + r\varepsilon,$$
$$\left.\begin{matrix}[Z_\varepsilon] = \text{space} \\ [\varepsilon] = \text{time}\end{matrix}\right\} \text{ hence } \begin{matrix}[r] = \text{velocity, and} \\ [\varepsilon r] = \text{space.}\end{matrix}$$

That is, the "modified" division

$$\varepsilon r = \frac{\varepsilon r \times p_\zeta}{1 \times p_\zeta} \quad \left[= \frac{\text{mass}}{\text{mass}}\right]$$

loses a dimension of space. This suggests to add one back, to the numerator. Conceptually this means that instead of as mass over 1D space $\big(\varepsilon r \times p_\zeta\big)$, we consider it as a box $\big(1 \times \varepsilon r \times p_\zeta\big)$ — a mass over 2D space. Having one extra spatial dimension in the domain of the law amounts to including the $Z_0 = \zeta$ variable and considering the *joint* law

$$\mathbb{P}|Z_\varepsilon, Z_0\rangle : \mathcal{B}(\mathbb{R}^2) \to [0, 1].$$

Let us see now how the time increment of $\mathbb{P}|Z_t, Z_0\rangle$ looks like with the last derivation in mind.

A.3.1 Diffusions

> **Context.** Consider the diffusion process
>
> $$Z_t = \zeta + rt + W_t$$
>
> for a constant $r > 0$ and an initial $Z_0 = \zeta$ independent from the Wiener process $(W_t)_t$.

Derivation. Similarly to the diffusion-free context, for any $a \in \mathbb{R}$ we have

$$\begin{aligned}\mathbb{P}(Z_t \le a) - \mathbb{P}(Z_0 \le a) \\ = -\mathbb{P}(a - r\,t - W_t \le \zeta \le a).\end{aligned}$$

In Equation 5 we expressed the probability as the length of the interval times the density of ζ at a point in it, but in this case, the interval is random — its length depends on W_t. We can still evaluate the joint law, though:

$$\begin{aligned}
\mathrm{d}\mathbb{P}|Z_0 = x, Z_t = z\rangle &= \mathrm{d}\mathbb{P}|\zeta = x, \zeta + r\,t + W_t = z\rangle \\
&= \mathrm{d}\mathbb{P}|\zeta = x, W_t = z - r\,t - x\rangle \\
&\quad (\text{by independence } \zeta \perp W_t)) \\
&= \mathrm{d}\mathbb{P}|\zeta = x\rangle \cdot \mathrm{d}\mathbb{P}|W_t = z - r\,t - x\rangle
\end{aligned}$$

We see the evolution dynamics integrated along the initial condition ζ. The Radon-Nikodym derivative above then corresponds to disintegrating this expression, i.e. taking the integrand only, which is a map

$$\begin{aligned}
&\mathbb{R} \times \mathcal{B}(\mathbb{R}) \to \mathbb{R} \\
&\quad (x, B) \mapsto \int_B 1_{A,B}(x, x + r\,t + y)\, \mathrm{d}\mathbb{P}|W_t = y\rangle
\end{aligned}$$

shaped precisely as a Markov kernel.

That is, the law $\mathbb{P}|\zeta\rangle$ naturally falls of from $\mathbb{P}|Z_0, Z_\varepsilon\rangle$, and we cancel ζ's effect completely via the Radon-Nikodym derivative:

$$\frac{\mathrm{d}\mathbb{P}|Z_0, Z_\varepsilon = y\rangle}{\mathrm{d}\mathbb{P}|Z_0\rangle}(x) = \mathrm{d}\mathbb{P}|W_t = z - r\,t - x\rangle.$$

On functions the joint law decomposition above acts by

$$\begin{aligned}
\iint f\, \mathrm{d}\mathbb{P}|Z_0, Z_t\rangle &= \iint f(x, z)\, \mathrm{d}\mathbb{P}|Z_0 = x, Z_t = z\rangle \\
&= \int \left(\int f(x, z)\, \mathrm{d}\mathbb{P}|W_t = z - r\,t - x\rangle \right) \mathrm{d}\mathbb{P}|\zeta = x\rangle \\
&\quad (\text{change variables } \; y \equiv z - r\,t - x) \\
&= \iint \underbrace{f(x, y + r\,t + x)\, \mathrm{d}\mathbb{P}|W_t = y\rangle}_{\text{pure evolution}}\, \mathrm{d}\mathbb{P}|\zeta = x\rangle,
\end{aligned}$$

□

Conclusion. While the law of Z_t contains the pure evolution increments in a non-trivial way, extractable (in the $Z_t = \zeta + r\,t$ case) by a Radon-Nikodym derivative against the initial condition's spatial derivative, the joint law of (Z_0, Z_t) contains the entire Z_0 dependence as a clean factor that is easily cancellable by a Radon-Nikodym derivative against $\mathbb{P}|Z_0\rangle$, resulting in

$$\frac{\mathrm{d}\mathbb{P}|Z_t, Z_0\rangle}{\mathrm{d}\mathbb{P}|Z_0\rangle} = \mathbb{P}|Z_t \mid Z_0\rangle.$$

B. Dirac Bra-Ket Notation for Conditional Laws

We will borrow Dirac's bra-ket notation and apply it to probability laws. The motivation came from a need for a saner looking formula for the Chapman-Kolmogorov equations. As this is not a standard presentation, we lay out the language and notation in this section.

Context. A probability space $(\Omega, \mathbb{P})$ and random variables $X, Y, Z : \Omega \to D$.

We first define the product of a random variable X and a probability measure $\mathbb{P}$ as the measure $X \cdot \mathbb{P}$ acting on sets by

$$\forall B \in \mathrm{dom}(\mathbb{P}) : (X \cdot \mathbb{P})(B) \coloneqq \int_B X \, \mathrm{d}\mathbb{P}.$$

Now we can consider the restriction of that measure to the σ-algebra generated by some random variable Y, denoted by $X \cdot \mathbb{P}\,|_Y$. Clearly if $\mathbb{P}$ vanishes on some set, then so does $X \cdot \mathbb{P}|_Y$ because multiplication of a finite number (X) with zero is still zero; that is, the absolute continuity condition

$$X \cdot \mathbb{P}|_Y \ll \mathbb{P}$$

is satisfied and so an almost-everywhere (in D) defined Radon-Nikodym derivative exists

$$\frac{\mathrm{d}X \cdot \mathbb{P}|_Y}{\mathrm{d}\mathbb{P}}(y) =: \mathbb{E}[X \mid Y = y] : D \multimap \mathbb{R},$$

called the conditional expectation.

By the relation $\mathbb{P}(B) = \mathbb{E}[1_B]$ it then induces a conditional probability

$$\forall A \in \mathcal{B} : \mathbb{P}(X \in A \mid Y) \coloneqq \mathbb{E}[1_{X \in A} \mid Y],$$

so $\mathbb{P}(X \in A \mid Y = y)$ is defined for almost every $y \in D$.

If we now let both $A \in \mathcal{B}(D)$ and $y \in D$ vary we get an object that we denote by

$$\begin{aligned} \mathbb{P}|X \mid Y \rangle\!\rangle : \mathcal{B}(D) \times D &\multimap [0,1] \\ (A, y) &\mapsto \mathbb{P}(X \in A \mid Y = y), \end{aligned}$$

where the y dependence is still *almost* sure.

So $\mathbb{P}|X \mid Y \rangle\!\rangle$ is the conditional law of X given Y. Let us then write the unconditional law of a variable Z as simply $\mathbb{P}|Z\rangle$.

As a positive function of $y \in D$, $\mathbb{P}(X \in A \mid Y = y)$ admits an integral against other measures, so it is well-defined to integrate it against the law of the variable Z:

$$\int \mathbb{P}|X \in A \mid Y = y \rangle\!\rangle \, \mathrm{d}\mathbb{P}|Z = y\rangle.$$

Let us denote this number by

$$\mathbb{P}|X \in A \mid Y = y \rangle\!\rangle \; Z = y\rangle,$$

where we treat y as a bound variable within the expression and write it only to make it clear what is the integrand (Y) and what is the integrator (Z). Alternatively, we can omit writing the variable y whatsoever and denote the same quantity by just

$$\mathbb{P}|X \in A \mid Y \rangle\!\rangle \; Z\rangle.$$

If we let $A \in \mathcal{B}$ vary, we get a back the conditional law of X given Y integrated against Z and denote it simply by

$$\mathbb{P}|X \mid Y \gg Z\rangle : \mathcal{B}(D) \to [0,1].$$

Chapman-Kolmogorov

> **Context.** A real-valued Markov process $(Y_t)_t$.

For two time points $t_1 \leq t_2$ and an $\varepsilon > 0$, the Chapman-Kolmogorov equation for the process Y states that the probability flux between times t_1 and $t_2 + \varepsilon$ goes through all possible values at the midtime t_2:

$$\mathbb{P}|Y_{t_2+\varepsilon} \mid Y_{t_1} \gg = \mathbb{P}|Y_{t_2+\varepsilon} \mid Y_{t_2} \gg Y_{t_2} \mid Y_{t_1} \gg .$$

In cases like this where all variables are instantiations of a process at different times, we take the liberty of writing $\mathbb{P}_Y \mid s \mid t : y \gg$ for $\mathbb{P}|Y_s \mid Y_t = y \gg$, so the last equality reads

$$\mathbb{P}_Y|t_2 + \varepsilon \mid t_1 \gg = \mathbb{P}_Y|t_2 + \varepsilon \mid t_2 \gg t_2 \mid t_1 \gg.$$

Now we are interested to derive the time-evolution of the left-hand side, and so we need only two time points. Fix $t > 0$ and $\varepsilon > 0$ and plug $t_1 = t_2 = t$ above:

$$\mathbb{P}_Y|t + \varepsilon \mid t \gg = \mathbb{P}_Y|t + \varepsilon \mid t \gg t \mid t \gg$$

and integrate along $\mathbb{P}|Y_t\rangle = \mathbb{P}|t\rangle$ to eliminate the trailing $|t\rangle$s:

> **Conclusion.** For any Markov $(Y_t)_t$ and times $t, t + \varepsilon$,
>
> $$\mathbb{P}_Y|t + \varepsilon\rangle = \mathbb{P}_Y|t + \varepsilon \mid t \gg t\rangle,$$
>
> or, more explicitly,
>
> $$\mathbb{P}|Y_{t+\varepsilon}\rangle = \mathbb{P}|Y_{t+\varepsilon} \mid Y_t \gg Y_t\rangle.$$

C. The Fokker-Planck Equation

> **Context.** While we define[14] a D-valued stochastic process $(Y_t)_t$ on a probability space $(\Omega, \mathbb{P})$ as a map of shape $Y : I \times \Omega \to D$ measurable in the Ω argument, what is *relevant* are all its laws $\mathbb{P}|Y_t\rangle, t \geq 0$, as well as all its joint laws
>
> $$\text{for all } n \text{ and } t_1, ..., t_n, \quad \mathbb{P}\Big|Y_{t_1}, ..., Y_{t_n}\Big\rangle : \mathcal{B}(D^n) \to [0,1].$$
>
> So from the perspective of time flowing, what is relevant is not so much the sequence $(Y_t)_t$ of the *variables*, but rather the sequence $\mathbb{P}|Y_t\rangle_{t\geq 0}$ of their *laws*.
>
> If Y admits probability densities $p(t,x) = \mathrm{d}\mathbb{P}|Y_t\rangle/dx$, it is natural to consider the sequence of densities $(p(t,\cdot)_t)_t$ and ask if their evolution is driven by a partial differential equation

[14] usually for an interval $I \subseteq \mathbb{R}_{\geq 0}$ and a Borel $D \subseteq \mathbb{R}^N$

$$\frac{\partial p(t,x)}{\partial t} = F[t, x, p, \partial_x p, \partial_{xx} p, ...], \text{for some } F,$$

and in particular, **what is that F?** In this section, we answer that question positively for Itô diffusions.

In Section A. we explained that the proper object to study the distributional evolution of an Itô stochastic process is not $\mathbb{P}|Y_t\rangle$ but $\mathbb{P}|Y_t \mid Y_0\rangle$, we present one approach to deriving the Fokker-Planck equation.

We will derive it by Tailor-expansion of the transition densities, following the general approach in (Daniel Walter, 2021) or the referenced inside (J. L. García-Palacios, 2004), sec. 4 and 5. However, we completely rework it in the language of measures and Markov kernels so that the probability distributions are not assumed to admit densities (against the Lebesgue measure, as done in these resources). This is why we will arrive at a PDE of distributions (weak form) and not of ordinary functions. Only after will we specialize to density-admitting processes (strong form).

C.1 Weak Form

The evolution

Abstract nonsense.

Context. We want to generalize the relation

$$\mathbb{P}[B] = \mathbb{E}[1_B], \quad \text{for every event } B,$$

from events B to random variables X, so that we can write the law of X as an expectation.

Derivation. On events regarding random variables, $B = (X \in A)$, the equivalence $\mathbb{P}[B] = \mathbb{E}[1_B]$ reads

$$\mathbb{P}(X \in A) = \mathbb{E}[1_{X \in A}]. \tag{7}$$

Below we are interested in using that equality over all $\mathcal{B}$orel A, in which case the left hand side can be seen as the law $\mathbb{P}|X\rangle$ evaluated at A. Let us write the right side's dependence on A as a measure, too.

First, realize that the randomness in the expectand in (7) is only through X, so we should be able to write $1_{X \in A}(\omega)$ as a function of $X(\omega)$. Indeed, $1_{X \in A} = 1_A(X)$.

Now recall that the characteristic function $(1_A(x))$ is the Dirac-delta function with flipped arguments: for all $\mathcal{B}$orel A and all $y \in \mathbb{R}$, $1_A(y) = \delta_y(A)$. Evaluated at $y := X$, this reads $1_A(X) = \delta_X(A)$. Chaining with the equality from the last paragraph, we get

$$1_{X \in A} = 1_A(X) = \delta_X(A) \tag{8}$$

and so

$$\forall\, \mathcal{B}\text{orel } A: \quad \mathbb{P}(X \in A) \overset{(7)}{=} \mathbb{E}[1_{X\in A}] \overset{(8)}{=} \mathbb{E}[\delta_X(A)],$$

which in point-free style reads

$$\mathbb{P}|X\rangle = \mathbb{E}[\delta_X]. \tag{9}$$

as an equality of measures. □

We will derive the Fokker-Planck equations by a Taylor expansion of the evolution of $\mathbb{P}|Y_t\rangle$.

Context. Fix a diffusion

$$\begin{aligned} dY_t &= b(t, Y_t)dt + \sigma(t, Y_t)dW_t \\ Y_0 &= y_0 \end{aligned}$$

with b, σ satisfying the relevant linear growth and local Lipschitz conditions so that a solution exists.

Derivation. We first rewrite

$$\mathbb{P}\big|Y_{t+\varepsilon} \;\big|\; Y_t \,\big\rangle\!\big\rangle \overset{(9)}{=} \mathbb{E}\Big[\delta_{Y_{t+\varepsilon}} \;\Big|\; Y_t\Big]. \tag{10}$$

Now we Taylor-expand the expectand in weak sense

$$\begin{aligned} \forall y' = y + \Delta: \quad \delta_{y'} &= \delta_{y+\Delta} \\ &= \sum_k \delta_y^{(k)} \frac{\Delta^k}{k!} \\ &= \delta_y + \delta_y' \Delta + \delta_y'' \frac{\Delta^2}{2} + \cdots \end{aligned}$$

Abstract nonsense. Since the main operation we want to perform — take time-derivatives — is not generally available for measures, we turn our measures into distributions in Schwarz sense. We indicate the Schwarz distribution corresponding to a probability law by dropping the probability sign, i.e. for each random variable X we define a distribution $|X\rangle$ via $\mathbb{P}|X\rangle$ simply by

$$\forall \varphi \in C_0^\infty: \quad \langle \varphi \mid X \rangle = \int \varphi(x)\, \mathrm{d}\mathbb{P}|X = x\rangle.$$

Similarly, if $\mathbb{P}|X \mid Y\rangle$ is a Markov kernel for two random variables X and Y, the induced conditional distribution is defined as the family of distributions indexed by the range of Y defined by

$$\langle \varphi \mid X \mid Y = y \rangle = \int \varphi(x)\, \mathrm{d}\mathbb{P}|X = x|Y = y\rangle,$$

again for all test functions $\varphi \in C_0^\infty$ with domain the range of X.

[15] by $\mathbb{E}\left[\delta_{Y_t}^{(k)} \;\middle|\; Y_t\right] = \delta_{Y_t}^{(k)}$ for all k.

which evaluated at $y = Y_t$ and $y' = Y_{t+\varepsilon}$ and substituted into Equation 10 yields[15]

$$\begin{aligned} \left|Y_{t+\varepsilon} \mid Y_t \right\rangle\!\!\rangle &= \delta_{Y_t} + \delta'_{Y_t}\mathbb{E}[\Delta Y_t \mid Y_t] \\ &\quad + \frac{1}{2}\delta''_{Y_t}\mathbb{E}\left[{\Delta Y_t}^2 \;\middle|\; Y_t\right] + \cdots \end{aligned} \tag{11}$$

Glossing over details, we evaluate the coefficients to be

$$\begin{aligned} &\mathbb{E}[Y_{t+\varepsilon} - Y_t \mid Y_t] \\ &= \mathbb{E}\left[\int_t^{t+\varepsilon} b ds + \int_t^{t+\varepsilon} \sigma dW_s \;\middle|\; Y_t\right] \\ &\left(\text{since } \mathbb{E}[\textstyle\int \sigma dW_s] = 0 \text{ and } \frac{1}{\varepsilon}\int_t^{t+\varepsilon} b ds \to b(t)\right) \\ &= b(Y_t, t)\cdot\varepsilon + o(\varepsilon) \end{aligned}$$

and

$$\begin{aligned} &\mathbb{E}\left[(Y_{t+\varepsilon} - Y_t)^2 \;\middle|\; Y_t\right] \\ &= \mathbb{E}\Big[\underbrace{\left(\textstyle\int b ds\right)^2}_{o(\varepsilon^2)} + \underbrace{\textstyle\int b ds}_{o(\varepsilon)}\underbrace{\textstyle\int \sigma dW}_{o(\sqrt{\varepsilon})} + \left(\textstyle\int \sigma dW\right)^2 \;\Big|\; Y_t\Big] \\ &= \mathbb{E}\left[\left(\textstyle\int_t^{t+\varepsilon} \sigma(s, Y_s) dW_s\right)^2 \;\middle|\; Y_t\right] + o(\varepsilon) \\ &\qquad \text{(Itô isometry)} \\ &= \mathbb{E}\left[\int_t^{t+\varepsilon} \sigma^2(s, Y_s) ds \;\middle|\; Y_t\right] + o(\varepsilon) \\ &= \sigma^2(t, Y_t)\cdot\varepsilon + o(\varepsilon), \end{aligned}$$

while higher-order coefficients consist of limits of expectations of linear combinations of terms

$$\begin{aligned} &\left(\int_t^{t+\varepsilon} b(t, Y_t) ds\right)^{k-i}\left(\int_t^{t+\varepsilon} \sigma(t, Y_t) dW_s\right)^i \\ &\qquad = o\big(\varepsilon^{k-i}\varepsilon^{i/2}\big) \subset o(\varepsilon) \quad (\text{for } \; k > 2). \end{aligned}$$

Then Equation 11 to the first order, i.e. with $\partial_\varepsilon|_{\varepsilon=0}$ applied, reads

$$\partial_\varepsilon \left|Y_{t+\varepsilon} \mid Y_t \right\rangle\!\!\rangle|_{\varepsilon=0} = b(t, Y_t)\delta'_{Y_t} + \frac{1}{2}\sigma^2(t, Y_t)\delta''_{Y_t},$$

or, introducing explicit variable for $Y_t = y$,

$$\partial_\varepsilon |t+\varepsilon \mid t : y \rangle\!\!\rangle|_{\varepsilon=0} = b(t, y)\delta'_y + \frac{1}{2}\sigma^2(t, y)\delta''_y,$$

to signify that this is an equality of distributions parametrized by y for almost every $y \in \mathbb{R}$, i.e. an equality of objects of shape $\mathbb{R} \multimap (C_0^\infty \to \mathbb{R})$.

Now we will integrate the free-hanging y against $\mathbb{P}|Y_t\rangle = \mathbb{P}_Y|t:y\rangle$, which on the left side turns

$$\mathbb{P}_Y|t+\varepsilon|t:y \rangle\!\rangle\ t:y\rangle = \mathbb{P}_Y|t+\varepsilon\rangle$$

and so we're left with an equation of distributions

$$\begin{aligned}\partial_\varepsilon|t+\varepsilon\rangle|_{\varepsilon=0} &= \int b(t,y)\delta'_y + \tfrac{1}{2}\sigma^2(t,y)\delta''_y \,\mathrm{d}\mathbb{P}|t:y\rangle. \\ &= \big|b(t,y)\delta'_y \rangle\!\rangle\ t:y\big\rangle \\ &\quad + \frac{1}{2}\ \big|\sigma^2(t,y)\delta''_y \rangle\!\rangle\ t:y\big\rangle.\end{aligned}$$

As distributions,

$$\begin{aligned}\big|b(t,y)\delta'_y \rangle\!\rangle\ t:y\big\rangle &= \partial|b(t,y)|t:y\rangle \\ \big|\sigma^2(t,y)\delta''_y \rangle\!\rangle\ t:y\big\rangle &= \partial^2\big|\sigma^2(t,y)|t:y\rangle,\end{aligned}$$

which can be easily derived[16] by acting on test functions. The right hand sides here are the distributional derivatives of the products of b and σ^2 with the distribution of Y_t, $|t:y\rangle$.

And so we derived the Fokker-Planck equation:

$$\begin{aligned}\partial_\varepsilon|t+\varepsilon\rangle|_{\varepsilon=0} &= \partial|b(t,y)|t:y\rangle \\ &\quad + \frac{1}{2}\partial^2\big|\sigma^2(t,y)|t:y\big\rangle\end{aligned}$$ □

We can spell the result in a less clumsy way if we leave the bound variable y implicit. For that we need to write the drift and the diffusion coefficients with the time argument as an index: $b_t(y), \sigma_t(y)$ instead of $b(t,y), \sigma(t,y)$.

Conclusion. Let $(Y_t)_t$ be the solution of the SDE

$$\begin{aligned}dY_t &= b_t(Y_t)dt + \sigma_t(Y_t)dW_t \\ Y_0 &= y_0\end{aligned}$$

where $b_t(y), \sigma_t(y)$ satisfy the linear growth and local Lipschitz conditions.

Then its laws $\mathbb{P}|Y_t\rangle$ satisfy the Fokker-Planck equation:

$$\partial_\varepsilon\big|Y_{t+\varepsilon}\rangle|_{\varepsilon=0} = \partial|b_t|Y_t\rangle + \frac{1}{2}\partial^2\big|\sigma_t^2|Y_t\big\rangle. \tag{12}$$

C.2 Strong Form

Now we specialize to the case when $(Y_t)_t$ has all densities.

[16] For any $\varphi \in C_0^\infty$,

$$\begin{aligned}\langle \varphi \mid b(t,y)\delta'_y \rangle\!\rangle\ t:y\rangle &= \langle b(t,y)\varphi \mid \delta'_y \rangle\!\rangle\ t:y\rangle \\ &= -\langle b(t,y)\dot\varphi(y) \mid t:y\rangle \\ &= -\langle \dot\varphi(y) \mid b(t,y)|t:y\rangle \\ &= \langle \varphi \mid \partial \mid b(t,y)|t:y\rangle\end{aligned}$$

Context. Let $(Y_t)_t$ be the solution of the SDE as above, and additionally assume that its laws admit densities against the Lebesgue measure, i.e.

$$\forall t \geq 0 : \quad \mathrm{d}\mathbb{P}|Y_t\rangle = p_t(y)dy$$

for functions p_t. We write the regular ket $|Y_t\rangle = |p_t\rangle$.

Put $|Y_t\rangle = |p_t\rangle$ in Equation 12 to get

$$\partial_\varepsilon |p_{t+\varepsilon}\rangle|_{\varepsilon=0} = \partial |b_t|p_t\rangle + \frac{1}{2}\partial^2 |\sigma_t^2|p_t\rangle.$$

For regular distributions we have $|b_t|p_t\rangle = |b_t p_t\rangle$ and $\partial|b_t p_t\rangle = -\,|\partial(b_t p_t)\rangle$, so the equation resolves to

$$|\partial_t p_t\rangle = -|\partial(b_t p_t)\rangle + \frac{1}{2}\,|\partial^2(\sigma_t^2 p_t)\rangle.$$

Equality of regular distributions is equality of the underlying functions, which gets us the classical PDE:

Conclusion. The Fokker-Planck equation on the densities p_t of $(Y_t)_t$:

$$\partial_t p_t = -\partial_y(b_t p_t)(y) + \frac{1}{2}\partial_y^2(\sigma_t^2 p_t)(y)$$

D. The $\nabla \log p$ Term

By general abstract theory ((U. G. Haussmann & E. Pardoux, 1986), as cited by (Montanari, 2025)), an Itô process

$$dX_t = b_t(X_t)dt + \sigma_t dW_t, \quad X_0 \sim \mu$$

has a time-reversal (under some assumptions on b, σ, μ)

$$dY_r = \left(-\overline{b}_t(Y_t) + \nabla_y \log p(t,y)\right)dt + \overline{\sigma}_t(Y_t)d\overline{W}_t$$

where $\overline{b}(t,y) = b(1-t,y)$ and $\overline{\sigma}(t,y) = \sigma(1-t,y)$, and $p(t,\cdot)$ is the density of X_{1-t}. While the negated b in the reversal makes sense (if X drifts down as time moves forward, then the backward Y must drift up), the logarithm gradient appears in a mysterious way.

Another source of $\nabla \log p$ terms is the equation of Langevin dynamics

$$dX_t = \nabla \log p(X_t) + \sqrt{2}dW_t, \text{for}$$

that serves as a sampling algorithm e.g. in (Yang Song & Stefano Ermon, n.d.). The $\nabla \log p$ term is called the *score* function, and a neural network is trained to learn it, so that later a discretisation of the Langevin equation can be run to produce samples.

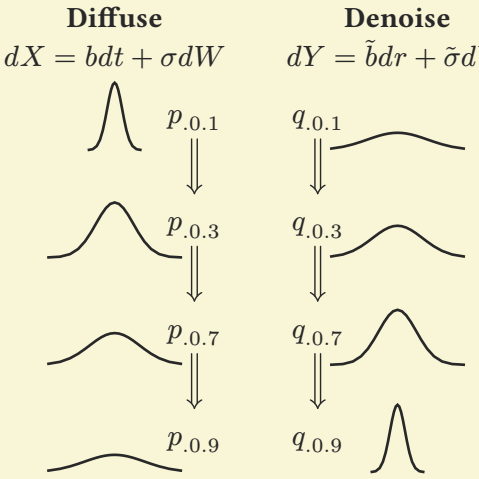
In this section we derive where exactly this $\nabla \log p$ comes from. We do not prove that the reversed process is a diffusion, we only derive what its coefficients must be if it exists.

D.1 For Time-Reversed Diffusions

Pick an SDE

$$dX_t = b_t(X_t)dt + \sigma_t dW_t, \quad X_0 \sim \mu$$

and put its distributions $p_t dx = d\mathbb{P}_{X_t}$.



Now let us find $\tilde{b}, \tilde{\sigma}$ that make for a process Y_r having the densities of X_t in reverse order.

So we are after $\tilde{b}, \tilde{\sigma}$ such that $dY_r = \tilde{b}_r(Y_r)dr + \tilde{\sigma}_r dW_r$ has densities $d\mathbb{P}_{Y_r} = d\mathbb{P}_{X_t}$.

Let $q_r dx = d\mathbb{P}_{Y_r}$. Then we desire $q_r = p_t$ for all $r \sim t$.

Both p_t, q_r satisfy the respective Fokker-Plancks

$$\dot{p_t} = F[b_t, \sigma_t, p_t] \qquad \text{for all } t \in I$$
$$\dot{q_r} = F\left[\tilde{b}_r, \tilde{\sigma}_r, q_r\right] \qquad \text{for all } r \in I$$

To connect the $\tilde{b}, \tilde{\sigma}$ to b, σ we need to couple the two equations. Clearly we can get a relationship between $\dot{p}$ and $\dot{q}$ if we ∂_r the equality $q_r = p_t$ (considering $r = r(t)$, we have $\dot{r}$):

$$\dot{q}_r = \partial_r q_r \overset{!}{=} \partial_r p_t = \dot{p}_t \partial_r t = \dot{t}\dot{p}_r.$$

Naturally, if we pick $r \sim t$ to be $r + t = T$, the relation is $\dot{q}_r = -\dot{p}_t$, meaning that p_t flows oppositely to q_t. Applying Fokker-Planck now gives

$$F\left[\tilde{b}_r, \tilde{\sigma}_r, q_r\right] = \dot{t}F[b_t, \sigma_t, p_t].$$
$$\nabla \cdot \left[\tilde{b}_r q_r\right] - \frac{1}{2}\Delta\left[\tilde{\sigma}_r^2 q_r\right] = \dot{t}\left(\nabla \cdot [b_t p_t] - \frac{1}{2}\Delta\left[\sigma_t^2 p_t\right]\right)$$

We have two terms of shape $\nabla \cdot (p_t \, \cdot)$ and two of shape $\Delta(p_t \, \cdot)$; rearrange

$$\nabla \cdot \left[p_t(\tilde{b}_r - \dot{t}b_t)\right] = \frac{1}{2}\Delta\left[p_t(\tilde{\sigma}_r^2 - \dot{t}\sigma_t^2)\right]$$

By recalling that $\Delta = \operatorname{div} \circ \operatorname{grad}$, pull a div from both sides:

$$\operatorname{div}\left[p_t(\tilde{b}_r - \dot{t}b_t)\right] = \operatorname{div}\left[\frac{1}{2}\operatorname{grad}\left[p_t(\tilde{\sigma}_r^2 - \dot{t}\sigma_t^2)\right]\right].$$

Put $s_t \equiv \tilde{\sigma}_r^2 - \dot{t}\sigma_t^2$ for short. Split cases $p_t = 0$ and $p_t > 0$ in the right hand side to pull out a p_t term

$$\nabla[p_t s_t] = \left(1_{p_t>0} + 1_{p_t=0}\right)\nabla[s_t p_t]$$

Assuming we pick $s_t > 0$, note that on $p_t = 0$ we have[17] $\nabla s_t p_t = 0$, so the $1_{p_t=0}$ term vanishes:

$$\begin{aligned}
&= p_t \frac{\nabla s_t p_t}{p_t} 1_{p_t>0} \\
&= p_t s_t 1_{p_t>0} \nabla[\ln s_t p_t]
\end{aligned}$$

The equation has one obvious solution for $\tilde{b}_r$ now if we drop the div and divide by p_t:

$$\tilde{b}_r = \dot{t} b_t + \frac{1}{2} s_t 1_{p_t>0} \nabla[\ln p_t s_t] \quad \text{for all } r \sim t.$$

For a saner appearance, pick $\tilde{\sigma}_r = \sigma_t$ and $r + t = T$ so that $\dot{t} = -1$ and $s_t = 2\sigma_t^2$, then

$$\tilde{b}_r = -b_t + \sigma_t^2 1_{p_t>0} \nabla[\ln \sigma_t^2 p_t].$$

This derivation works if σ_t is given strictly positive for all times everywhere. If σ_t hits zero, the world shatters.

If $\sigma_t(x) = \sigma_t$ is constant in space, then[18] $\nabla[\ln \sigma_t^2 p_t] = \nabla[\ln p_t]$ and

$$\tilde{b}_r = -b_t + \sigma_t^2 \nabla[\ln p_t] 1_{p_t>0}.$$

III Sampling by poisson point processes

We now move our attention to sampling algorithms based on a Poisson point process.

Let x be a non-negative[19] real random variable.

A. Algorithm Derivation

(Montanari, 2025) states the algorithm, here show how to derive it.

> **Context.** Let $(Y_t)_t$ be a process that conditionally on x is distributed as a Poisson point process of rate x, in the sense described in Equation 1 on page 11. We denote this predicate from now on by
>
> $$(Y|x) \sim \mathrm{PPP}(x).$$
>
> This means that the jump probabilities of Y are required to satisfy
>
> $$\begin{aligned}
> \forall t \geq 0, \delta > 0 : \mathbb{P}(Y_{t+\delta} = Y_t + 1 \mid x) &= \delta x + o(\delta) \\
> \text{and } \mathbb{P}(Y_{t+\delta} = Y_t \mid x) &= 1 - \delta x + o(\delta),
> \end{aligned} \tag{13}$$

[17] p_t is non-negative, so at $x \in D$ with $p_t(x) = 0$, the gradient must be zero, otherwise $p_t(x - \varepsilon \nabla p_t(x)) < 0$ for sufficiently small $\varepsilon > 0$

[18] $\nabla[\ln \sigma_t^2 p_t] = \nabla[\ln \sigma_t^2 + \ln p_t] = \cancel{\nabla[\ln \sigma_t^2]} + \nabla[\ln p_t] = \nabla[\ln p_t]$

[19] so that it makes sense as a rate of a point process; we show a slight generalization of this later.

> i.e. the probability of a tick occurring (a jump of the counting variable Y_t) in any time interval $(t, t+\delta)$ is δx to the first order in δ.

Derivation. Note that these are equations of almost-everywhere defined functions of x, since we condition on x in the probabilities. As time progresses from $t = 0$ on, though, treating x as latent (not measurable at t), we only observe the history of Y_t; i.e. as far as the performer of the algorithm is concerned, the jump probabilities he or she observes are the expected values of Equation 13 given Y_t:

$$\mathbb{E}[\mathbb{P}(Y_{t+\delta} = Y_t + 1 \mid x) \mid Y_t] = \delta\, \mathbb{E}[x \mid Y_t] + o(\delta).$$

Denote the jump event by $\hat{J}_{t,\delta} := (Y_{t+\delta} = Y_t + 1)$. Now, the σ-algebra generated by x is no rougher than Y_t at any time, so expected probability of the event of a jump $\hat{J}_{t,\delta}$ on the left is simply

$$\begin{aligned}\mathbb{E}\Big[\mathbb{P}\Big(\hat{J}_{t,\delta} \mid x\Big) \;\Big|\; Y_t\Big] &= \mathbb{E}\Big[\mathbb{E}\Big[1_{\hat{J}_{t,\delta}} \mid x\Big] \mid Y_t\Big] \\ &\overset{!}{=} \mathbb{E}\Big[1_{\hat{J}_{t,\delta}} \mid Y_t\Big] = \mathbb{P}\Big(\hat{J}_{t,\delta} \mid Y_t\Big),\end{aligned}$$

so the transition probabilities of Y_t are

$$\begin{aligned}\mathbb{P}(Y_{t+\delta} = Y_t + 1 \mid Y_t) &= \delta\, \mathbb{E}[x \mid Y_t] + o(\delta) \\ \text{and } \mathbb{P}(Y_{t+\delta} = Y_t \mid Y_t) &= 1 - \delta\, \mathbb{E}[x \mid Y_t] + o(\delta).\end{aligned}$$

□

Putting $m(t, y) = \mathbb{E}[x \mid Y_t = y]$ (for each t, an almost surely defined function of y), this gives

Conclusion. An algorithm for sampling Y:

Discretise time into $t_0 = 0, \delta, 2\delta, \ldots, M\delta$. Iterating over k:

At $t = 0$,	let $y := 0$.
At each $t = k\delta$,	increment $y \leftarrow (y + 1)$ with probability $\delta \cdot m(t, Y_t)$, otherwise keep Y_t the same.

The rate x of Y_t that the algorithm recovers this way at time T, i.e. the sample x that we output, is then

$$x = \frac{y}{T}.$$

Of course, here we need to be able to evaluate $m(t, y)$, as we mentioned in the first section.

B. Correctness

> **Context.** Let again $x \geq 0$ and $(Y \mid x) \sim \mathrm{PPP}(x)$. Now we prove
>
> $$\lim_t \frac{Y_t}{t} = x \text{ (almost surely)}.$$

Derivation. The counting variable Y_t of a ticking (point) process is the number[20] of inter-arrival times $\xi_i, i \in \mathbb{N}$ that occur up to time t:

$$Y_t = \#_i(\xi_i < t),$$

and the nth arrival time is the sum

$$T_n = \sum_{i=1}^{n} \xi_i.$$

Any time $t \geq 0$ then lies between two consecutive arrival times[21]:

$$\forall t, n : \quad \sum_{n \leq Y_t} \xi_n \leq t < \sum_{n \leq Y_t + 1} \xi_n. \tag{14}$$

For our poisson point proccess all ξ_i are independent and exponentially distributed of parameter x. By the law or large numbers then the average of ξ_i converges almost surely to the mean, $1/x$. That is,

$$\lim_t \frac{\sum_{n \leq Y_t} \xi_n}{Y_t} = \frac{1}{x}. \tag{15}$$

Divide Equation 14 by Y_t then to squeeze $\frac{t}{Y_t}$ inside $\frac{1}{x}$:

$$\begin{aligned} \frac{\sum_{n \leq Y_t} \xi_n}{Y_t} \leq \frac{t}{Y_t} &< \frac{\sum_{n \leq Y_t + 1} \xi_n}{Y_t} \\ &= \frac{\sum_{n \leq Y_t + 1} \xi_n}{Y_t + 1} \cdot \frac{Y_t + 1}{Y_t}, \end{aligned}$$

apply the squeeze lemma and Equation 15 to the limit as $t \to \infty$:

$$\frac{1}{x} \leq \lim_t \frac{t}{Y_t} \leq \frac{1}{x},$$

proving that

$$\frac{Y_t}{t} = x. \qquad \square$$

C. A Speedup Of $g(t)$

Here we're generally interested in what happens if instead of running a process $(Y \mid x) \sim \mathrm{PPP}(x)$ of rate x, we use a rate[22] $f(t, x)$ and run $\left(Y^{(f)} \mid x\right) \sim \mathrm{PPP}(t \mapsto f(x, t))$.

> **Context.** We first consider the special case $f(t, x) = g(t)\, x$ and a process

[20] If for $i \in \mathbb{N}$, A_i are events, we put $\#_i A_i$ to be the random variable counting how many of them occured: $\#_i A_i := \sum_i 1_{A_i}$

[21] as long as the process does not stop in finite time.

[22] again constrained nonnegative

$$\left(Y^{(g)} \mid x\right) \sim \mathrm{PPP}(t \mapsto g(t)\, x),$$

for a positive function $g(t) > 0$.

At each $\omega \in \Omega$, $Y(\omega)$ behaves like a Poisson point process of rate $g(t)x(\omega)$. Compared to our canonical $(Y \mid x) \sim \mathrm{PPP}(x)$, this means that a global (over x's values) time-speedup of $g(t)$ has been applied.

What does this mean in terms of our sampling algorithm? Can it still produce a sample of x if it runs $Y^{(g)}$ instead of Y?

Derivation. For the canonical Y with rate x, Y_t is a Poisson random variable of rate $t\,x$. The modified $Y^{(g)}$ consists of Poisson random variables of rate $\int_0^t g(s)xds$:

$$(Y_t^{(g)} \mid x) \sim \mathrm{Poisson}(G(t)\, x), \quad \text{for } G(t) = \int_0^t g(s)ds$$

If we can find a change of time $\tau = \tau(t)$ such that $Y^{(g)}_{\tau(t)}$ is distributed as a Poisson variable of rate $t\,x$, so that $(Y^{(g)}_{\tau(t)})_t \mid x \sim \mathrm{PPP}(x)$, then the correctness proof applies and $Y^{(g)}_{\tau(t)}/t$ will approach the sample x.

Indeed, to find a time τ such that the Poisson parameter $G(t)\,x$ turns out to be $t\,x$, means to find τ such that $G(\tau(t)) = t$. In other words - τ must be an inverse of G. As an integral of a positive function g, G is continuous and monotonic and hence invertible on its range $[0, G(T)]$, as long as T is picked so that $G(T)$ is finite.

Now assume $G(t)$ is invertible with inverse τ, so that

$$\left(Y^{(g)}_{\tau(t)} \mid x\right) \sim \mathrm{Poisson}(tx)$$

and the rescaled back process $(Y^{(g)}_{\tau(t)})_t$ is[23] a Poisson point process of rate x. Then by the correctness proof, it induces a sample-approximating sequence:

$$\frac{Y^{(g)}_{\tau(t)}}{t} \xrightarrow[t\to\infty]{} x.$$

However, we are interested in the behaviour not of the normalised $(Y^{(g)}_{\tau(t)})_t$, but of the sped-up $Y^{(g)}$. To get that, we change variables, $t = G(\tau(t))$:

$$x \xleftarrow[t\to\infty]{} \frac{Y^{(g)}_{\tau(t)}}{t} = \frac{Y^{(g)}_{\tau(t)}}{G(\tau(t))},$$

$$\text{and so } \frac{Y^{(g)}_{\tau}}{G(\tau)} \xrightarrow[\tau\to\infty]{} x \qquad \square$$

Conclusion. If g is a positive integrable function of time, then a sample x can be approximated by

[23] we do not prove that the time-rescaled Poisson variables $Y^{(g)}_{\tau}$ indeed induce a Poisson point process

$$\frac{Y_\tau^{(g)}}{G(\tau)}, \quad \text{where} \quad \begin{array}{l} \left(Y^{(g)} \mid x\right) \sim \mathrm{PPP}(t \mapsto g(t)x) \\ \text{and } G(\tau) = \int_0^\tau g(t)dt \end{array}$$

D. Speedup Measurement

The speedup factor g discussed above makes the point process tick $g(t)$ times more often. This makes us expect that there will be an improvement in the approximation $Y_\tau^{(g)}/G(\tau)$ at a fixed time T.

> **Context.** Fix again a $g > 0$, put $G(t) = \int_0^t g$ and pick a boosted process
>
> $$(Y^{(g)} \mid x) \sim \mathrm{PPP}(t \mapsto g(t)x).$$
>
> Pick $T > 0$ and define the Prokhorov distance between the laws of the target x and the approximating sample $X_T := Y_T^{(g)}/G(T)$ to be the infimum of
>
> $$\left\{ \varepsilon > 0 \;\middle|\; \forall A \in \mathcal{B} : \begin{array}{c} \mathbb{P}(X_T \in A) \leq \mathbb{P}(x \in A^\varepsilon) + \varepsilon \\ \text{and } \mathbb{P}(x \in A) \leq \mathbb{P}(X_T \in A^\varepsilon) + \varepsilon \end{array} \right\},$$
>
> and A^ε denotes the ε-expansion[24] of any $A \in \mathcal{B}(\mathbb{R})$.
>
> $\pi_T := \inf P_T$, where P_T denotes the set
>
> The question is, can we bound π_T?

Derivation. In (Patrick Billingsley, 1999) we find a statement of a theorem by Strassen-Dudley, whose converse allows us to rework the Prohorov distance π_T. In particular, we consider the following decomposition of the event $X_T \in A$:

$$X_T \in A \subseteq (|x - X_T| > \varepsilon) \cup (x \in A^\varepsilon).$$

It is easiest to visualise what this means if we draw the product space $\mathbb{R} \otimes \mathbb{R}$ and treat the two axes as the axes of values of the random variables x and X_T; then the inclusion above is a statement about areas in $\mathbb{R}^2$.

The inclusion implies a respective inequality about probabilities,

$$\mathbb{P}(X_T \in A) \leq \mathbb{P}(|x - X_T| > \varepsilon) + \mathbb{P}(x \in A^\varepsilon). \tag{16}$$

Now *if* we find an $\varepsilon > 0$ that keeps the left summand less than ε, so satisfies

$$\mathbb{P}(|x - X_T| > \varepsilon) < \varepsilon, \tag{17}$$

we will get the predicate in the infinmum defining π_T:

$$\mathbb{P}(X_T \in A) \leq \varepsilon + \mathbb{P}(x \in A^\varepsilon). \tag{18}$$

Thus in our goal of bounding π_T, where we have to find an $\varepsilon > 0$ satisfying Equation 18 (the smaller, the better), we can search instead for $\varepsilon > 0$ satisfying simply Equation 17.

Equation 17, however, means

[24] $A^\varepsilon := \{y \mid \exists y' \in A : |y - y'| < \varepsilon\}$

$$\mathbb{P}\Big(|x - Y_T^{(g)}/G(T)| > \varepsilon\Big) < \varepsilon,$$

and we can condition on x to apply Poisson'ness of $Y_T^{(g)}$:

$$\begin{aligned}
&\mathbb{P}\Big(|x - Y_T^{(g)}/G(T)| > \varepsilon\Big) \\
&= \mathbb{E}\Big[\mathbb{P}\Big(|x - Y_T^{(g)}/G(T)| > \varepsilon \;\Big|\; x\Big)\Big] \\
&\text{(by the double-sided Chernoff bound[25])} \\
&\le 2\,\mathbb{E}[\exp(-G(T)\,h(\varepsilon/x)\,x)] \\
&\qquad \text{for } h(u) = (1+u)\ln(1+u) \\
&\le 2\,\mathbb{E}[\exp\left(-\frac{G(T)\varepsilon^2}{2\big(x + \frac{\varepsilon}{3}\big)}\right)
\end{aligned}$$

□

Conclusion. As a bound on the distance π_T we can take the solution of the equation

$$2\varepsilon = \mathbb{E}\left[\exp\left(-\frac{G(T)\varepsilon^2}{2\big(x + \frac{\varepsilon}{3}\big)}\right)\right],$$

which cannot be solved analytically in general.

E. If x Is Not Nonnegative

The rate of a point process naturally must be nonnegative, but we can also ask if this framework is still useful for some not necessarily nonnegative random variables x. For example, if we decompose $x = \operatorname{sign}(x) \cdot |x|$, we can apply the machinery above with $|x|$. Then, if $\operatorname{sign}(x)$ and $|x|$ are independent, we can sample separately $|x|$ by the algorithm above, sample $\epsilon \sim \operatorname{sign}(x)$, and then emit $\epsilon\, Y_T/T$. A natural question then is, under what conditions are $\operatorname{sign}(x)$ and $|x|$ independent?

Turns out, we can prove the following equivalence:

$$\begin{aligned}
&(|x| \perp \operatorname{sign}(x) \mid x \neq 0) && \text{(I)} \\
\Leftrightarrow\ &\mathbb{P}(x \mid x > 0) = \mathbb{P}(-x \mid x < 0), && \text{(L)}
\end{aligned}$$

which in words reads as the equivalence of the following:

1. (I) Given $x \neq 0$, $|x|$ and $\operatorname{sign}(x)$ are independent;
2. (L) The conditional law of x given $(x > 0)$ is equal to the conditional law of $-x$ given $(x < 0)$.

That is, the sign is independent from the modulus exactly when x's law, normalised on $x > 0$, coincides with its negative's law normalised on $x < 0$. We condition on $x \neq 0$ to allow cases with non-zero probability for the event $(x = 0)$.

[25] the only place in this thesis we have not thoroughly verified an LLM's output. Hopefully I will find time to write this out in full detail in the near future.

Proof of ($\Longrightarrow$). The forward direction is direct, as the following chain of equalities of the laws of $x, -x, |x|$ shows:

$$\begin{aligned} \mathbb{P}(x \mid x > 0) &= \mathbb{P}(|x| \mid x > 0) \\ &\overset{\perp}{=} P(|x| \mid x \neq 0) \overset{\perp}{=} \mathbb{P}(|x| \mid x < 0) && \text{(Assumed (I))} \\ &= \mathbb{P}(-x \mid x < 0). \end{aligned}$$

□

Proof of ($\Longleftarrow$). The converse relies on Bayes' formula.

Assuming temporarily that $(x = 0)$ happens almost never, and the equality of laws (L), we have that for both $\epsilon \in \{-1, +1\}$ and all $\mathcal{B}$orel A,

$$\begin{aligned} \mathbb{P}(|x| \in A) &= \mathbb{P}(\epsilon x \in A, \epsilon x > 0) + \mathbb{P}(-\epsilon x \in A, \epsilon x < 0) \\ &= \mathbb{P}(\epsilon x \in A \mid \epsilon x > 0)\, \mathbb{P}(\epsilon x > 0) && \text{(Bayes)} \\ &\quad + \mathbb{P}(-\epsilon x \in A \mid \epsilon x < 0)\, \mathbb{P}(\epsilon x < 0) \\ &= \mathbb{P}(\epsilon x \in A \mid \epsilon x > 0)\, \mathbb{P}(\epsilon x > 0) \\ &\quad + \mathbb{P}(\epsilon x \in A \mid \epsilon x > 0)\, \mathbb{P}(\epsilon x < 0) && \text{(L)} \\ &= \mathbb{P}(\epsilon x \in A \mid \epsilon x > 0)[\mathbb{P}(\epsilon x > 0) + \mathbb{P}(\epsilon x < 0)] && \text{(grouping)} \\ &= \mathbb{P}(\epsilon x \in A \mid \epsilon x > 0) && (x \neq 0) \\ &= \mathbb{P}(|x| \in A \mid \operatorname{sign}(x) = \epsilon). \end{aligned}$$

Without the assumption that $(x = 0)$ almost doesn't happen, the equality chain still holds if we condition everything on $(x \neq 0)$[26]:

$$\mathbb{P}(|x| \in A \mid x \neq 0) = \mathbb{P}(|x| \in A \mid \operatorname{sign}(x) = \epsilon).$$

But this means exactly that the conditional law of $|x|$ (on $\operatorname{sign}(x)$) equals the unconditional, i.e. $|x|$ is independent from $\operatorname{sign}(x)$. □

Hence, if x has such a symmetric law, we can still use the Poisson point process based algorithm to sample it (if we can also sample its sign).

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[26] and observe that $(\operatorname{sign}(x) = \epsilon) \subset (x \neq 0)$, because $\epsilon \neq 0$.

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